

ViCurriAS: A CURRICULUM VISUALIZATION TOOL FOR FACULTY, ADVISORS, AND STUDENTS*

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ABSTRACT

This paper introduces ViCurriAS, a visual tool to aid curriculum committees, faculty, and advisors in mapping a particular program of study. ViCurriAS also allows advisors and students to see the course dependencies and progress for the aforementioned program of study. ViCurriAS consists of two integrated environments: one for developing and arranging the flow of courses for a particular program or to aid in the introduction of new courses to an existing curriculum; and the second to provide an advising tool to visually show student progress through the program.

1. INTRODUCTION AND PURPOSE

Have you ever tried to visualize the flow of course offerings when a new course or courses were being introduced? Have you ever tried to convince a student that certain courses should be taken early to avoid prerequisite problems in the future? Have you found it difficult to keep track of the many different concentrations offered by your department? Computing degree programs offer sets of courses, each following a path of pre, post and co-requisites, using the outcomes from one course to build the student's maturity and skills to proceed to the next course. For example, Appendix B of the Computing Curricula 2001 provides 47 model course descriptions that are "best regarded as a common starting point for experimentation" [1]. Included in these descriptions are the course prerequisites. Despite, or perhaps because of, the detail of these course offerings, it is difficult to grasp the overall flow of the curriculum.

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Various tools have been used to assist in visualizing these course dependencies. For example, Office Visio 2007® can be used to develop a flowchart of the course offerings or Project 2007® can be used to draw a critical path showing the course dependencies. While each of these tools can help produce a usable map of the curriculum, these maps are just that, maps. The data can only be used to illustrate the course offering but the actual course information concerning the ordering of courses must be transcribed to paper, usually in checklist format. Another scenario may be that the checklist is developed first and then the picture drawn to help visualize the flow.

Often the curriculum is modified to make way for new course offerings. Included with the new or modified course description will be a list of course dependencies in the form of prerequisites and post-requisites.

The transcribing of information from checklist to flowchart or flowchart to checklist is prone to error. ViCurriAS (Visual Curriculum Advising System) integrates the visual and textual information for both curriculum mapping and advising, eliminating any missteps from curriculum mapping to advising.

2. PREVIOUS WORK

A great deal of work has been done in developing curricula and a great deal of work has been done in using software to visualize information. However, little has been done in the area of curricula visualization, either as tool for helping to layout or establish a curriculum or as an advising tool. Geroimenko and Chen [2] stated information visualization utilizes graphical images of an abstract information structure for human users. Curricula located on the web are usually presented in textual form (e.g., lists of course codes, titles, and description) which makes it difficult for potential students to understand [4]. Sommaruga and Catenazzi used extensible 3D graphics (X3D) to formalize their graphical representation of the curricula. While Sommaruga and Catenazzi's work produced multiple graphical representations of existing curricula, it did not appear to be an interactive tool for manipulating or arranging curricula. The work of laying out the information was derived from an Oracle database, which presumably was populated prior to creating the graph. That is, the graph was created from static information. Gestwicki [3] is developing a curriculum visualization tool, called CurricVis, that creates diagrams based on a set of Prolog knowledge base entries.

What if the graph itself could be used to layout a curriculum and also provide the textual data? This is what ViCurriAS is designed to do.

3. OVERVIEW

ViCurriAS is composed of two related modules. The first module, called Curriculum, is used in the development/modification stage of the curriculum. Curriculum allows faculty/advisors to arrange the courses in the curriculum by prerequisites. The second module, called Advising, allows the advisor to enter grades or scheduling information into the same chart that was created by the Curriculum module. Both modules break up the curriculum into a series of tabs including core courses, major electives, free electives, and general education. The general education tab is further

subdivided into areas of general education such as Writing, Literature, Mathematics, Sciences, etc.

3.1 ViCurriAS: Curriculum Module

This module enables faculty members and/or advisors to construct a curriculum map (see Figure 1). This can be accomplished in two ways: 1) inserting new course[s] into the curriculum and filling out the course information (e.g., course code, course name, credit hours, course characteristics, etc); 2) importing the courses from a file containing the aforementioned data. The imported courses are simply drawn on the screen, one course next to another. The courses must be manually placed in the correct position via drag and drop.

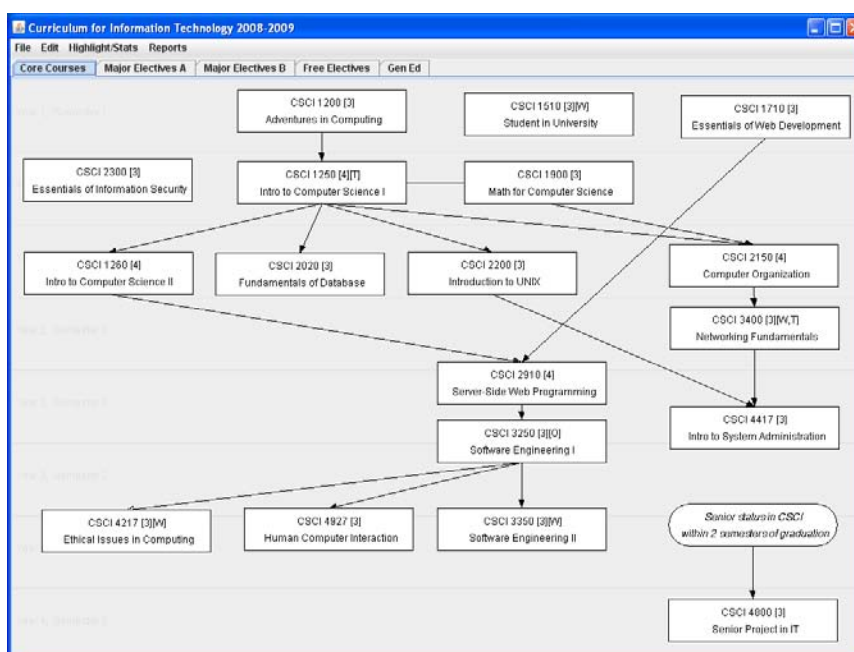


Figure 1. Sample Curriculum screenshot (with nodes suppressed)

The Curriculum module allows the faculty/advisors to connect the courses together, via lines called connectors, as prerequisites or co-requisites. In addition, users may add comment boxes to indicate a comment about one or more classes or about this particular portion of the curriculum. Once the Curriculum is set, it can be saved, printed as a diagram, or printed as a textual checklist.

3.2 ViCurriAS: Advising Module

The Advising module is based on the curriculum developed in the Curriculum module (see Figure 2).

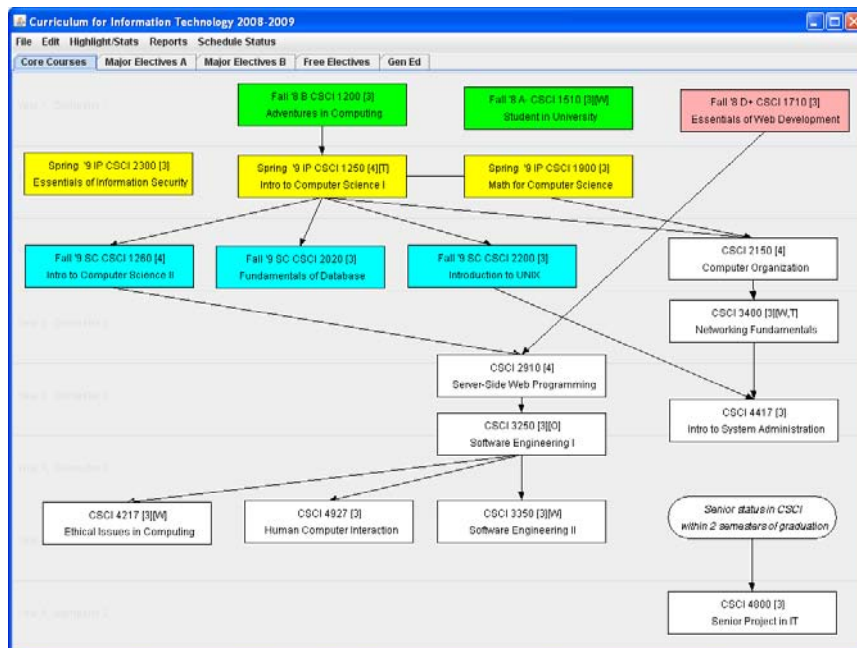


Figure 2. Sample Advising screenshot

When a student requires advising for the first time, the advisor opens up the proper curriculum map. In the Advising module the curriculum map cannot be altered, however, information regarding the status of the course (either a grade, or whether the class is scheduled or in progress) may be added as the student progresses through the curriculum. The status is color coded enabling the advisor and student to easily see his/her progress. The color coding is as follows: green courses were taken with a passing grade; pink courses were taken with an unsatisfactory grade; yellow courses are in progress in the current semester; and cyan courses are scheduled for the upcoming semester.

4. WORKING WITH ViCurriAS

4.1 Curriculum Module

In order to map the curriculum, three basic symbols were identified: course boxes, comment boxes, and connectors. Course boxes were further subdivided into three categories (see Table 1): a fixed defined course; a course with a fixed course code, but not a fixed title (e.g. Special Topics) or hours; and finally, a free elective course where the course code, course title, and hours are variable. Course boxes contain nodes on the top, bottom and both sides (see Figure 3). The top and bottom nodes are to indicate pre and post requisites and the side nodes are for connecting co-requisites. Data is entered into the course box via a course dialog (see Figure 4).

Table 1. Categories of Courses

Course Category	Example	How Defined
Fixed Course	Most courses fit this category, for	Course code and title are provided

Code, Title and Credit Hours	example: CSCI 1250 Intro to Computer Science	by user in appropriate boxes
Fixed Course Code, Variable Title and Credit Hours	Special topics courses, independent studies, etc.	Title Check box is checked course code and title are provided by user, title and/or hours may be changed during advising
Variable Course Code, Title, and Credit Hours	Free electives of the students choosing	Free Elective checkbox checked, course code and title are given as free elective. Course code, title, and hours will be determined in the advising module

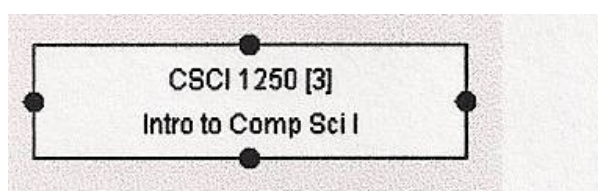


Figure 3. Course box with nodes

Figure 4. Course Entry Dialog

Comment boxes may contain up to two lines of comments about a specific course or about the program of study in general. Comments for a specific course are connected in the same way as pre and post requisites. The comment box is a rounded rectangle while the course box is a rectangle.

Connections to related courses are provided via connectors. There are two categories of connectors: directed and non-directed. Directed connectors are created using the convention that a prerequisite's bottom node is connected to a post requisite's top node or vice-versa. The co-requisite connector is not directed as no sequence is implied for co-requisites. All connectors are made by dragging the mouse from one course node to another.

The courses may be arranged by dragging and dropping using the mouse and the connectors will remain connected via rubber banding. If a course is deleted, all connectors to that course will also be deleted. A connector may be deleted individually.

Once the curriculum has been laid out, it may be saved to be used by the Advising module or used as a basis for further changes or other tracks. The user may also request a textual version of the curriculum and hard copy printouts of the diagram.

4.2 Advising Module

The Advising Module is designed to use the curricula developed in the Curriculum module as a template for advising. When a student arrives for an advising session, the advisor will determine if this is a new student or an existing student. If the student is new, the advising begins by selecting New Student from the file menu. This generates the curriculum map for his/her program of study. The advisor selects the courses, the student should be taking in the next semester, by left-clicking on a given course which opens up an Advising Dialog (see Figure 5). The advisor may enter the semester, year, and status. If the course category has a variable title, the title and actual credit hours should be entered at this time. If the course is a free elective, then the course code, name, and credit hours should be entered at this time. The advisor can only put in advising data, he/she may not alter the curriculum map.

If a student is a transfer student, it is possible to import his/her grades from a flat file containing the student's course code, grade, year and semester. The current format for the file is the above information separated by pound signs (#) because commas may appear in the course names. Obviously the format of this file may be modified; however, minor modifications to the software, to enable the file to be read and interpreted, must be made.

Once a student has developed a history, his/her information can be stored easily by saving the program of study (POS). This information can be accessed by clicking on the existing student menu and selecting the POS for that particular student. This will cause the advising module to read in the student's, previously saved, POS object.

Output from the Advising module includes: a diagram with color coded data (see Figure 2); a partially filled in textual POS; a chronological list of courses taken, in progress, and scheduled; and a list of scheduled classes.

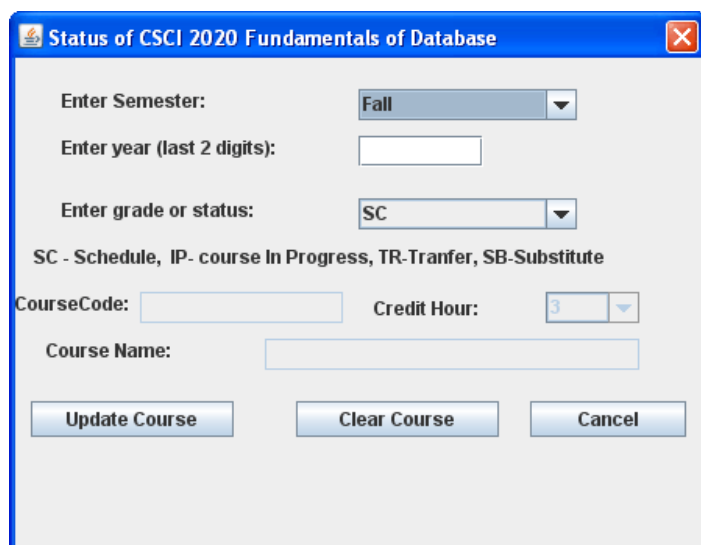
A screenshot of a Windows-style dialog box titled "Status of CSCI 2020 Fundamentals of Database". The dialog box has a blue title bar with a close button (X) in the top right corner. Inside, there are several input fields and buttons. The first row has "Enter Semester:" followed by a dropdown menu showing "Fall". The second row has "Enter year (last 2 digits):" followed by an empty text box. The third row has "Enter grade or status:" followed by a dropdown menu showing "SC". Below this, there is a legend: "SC - Schedule, IP- course In Progress, TR-Tranfer, SB-Substitute". The next row has "CourseCode:" followed by an empty text box and "Credit Hour:" followed by a dropdown menu showing "3". The final row has "Course Name:" followed by an empty text box. At the bottom, there are three buttons: "Update Course", "Clear Course", and "Cancel".

Figure 5. Advising Dialog

6. CONCLUSION

ViCurriAS, as a working prototype, enables users to develop a curriculum map to assist in the understanding of coursework flow. This visualization strategy provides added value by allowing users a way to arrange a series of courses into a coherent, sequenced, offering that allows the student to build on previous coursework and maturity. By saving the original design of the curricula, and using that design as a roadmap to completion for a given program of study, miscommunications during advisement may be reduced.

7. FUTURE WORK

Anecdotal comments indicate that students found the visual information very helpful in understanding the flow of courses required for graduation. Many students requested a copy of their advising diagram in order for them to begin planning the rest of their coursework. A more formal study should be done to indicate the degree of acceptance and usefulness of ViCurriAS.

While ViCurriAS is an effective tool in its current state, integration with other academic systems, such as Banner, can extend and improve the usability and accuracy of the maps. For example: A student may have been advised to take a course, but chose not to take the course due to scheduling problems, the course being full, etc. Another useful feature would be to highlight the course offerings for the intended semester so that the advisor and student will be able to schedule only the courses that are actually being offered. This will require importing the semester schedules into ViCurriAS.

8. REFERENCES

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